

# Embrace Edge AI, Empower Industrial Digitalization



## EC954 Series High-Performance AI Edge Computer

· Scalable AI Power

· High Security

· Rich Interfaces

· Cloud-Managed

# 1. Product Overview

EC954 is a high-performance multi-interface edge AI computer designed for industrial scenarios requiring edge intelligence, high reliability, and secure cloud integration.

Key features:

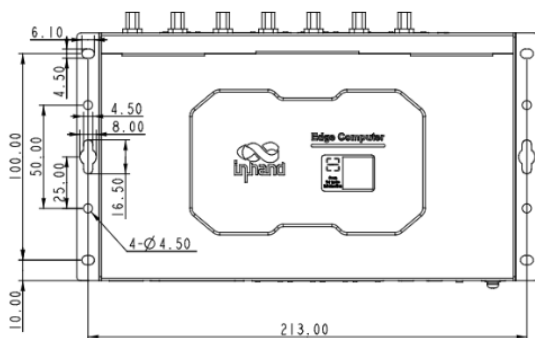
- **Scalable AI performance:** RKNN 1.0 TOPS, expandable up to 26 TOPS with Hailo-8 module
- **Rich industrial interfaces:** 4×GE, 12 serial channels, CAN, DI/DO, HDMI, USB, mSATA
- **Reliable connectivity:** Wired/cellular/Wi-Fi backup with dual-SIM failover
- **Security architecture:** Secure Boot, TPM2.0, TrustZone, firewall, VPN
- **Cloud operations:** DeviceLive remote monitoring and edge app/container management

## Core Technical Specifications

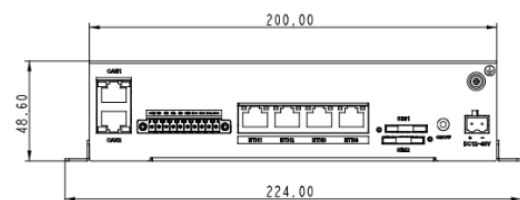
| Technical Indicator | Specification                                            |
|---------------------|----------------------------------------------------------|
| Cellular Network    | LTE Cat.4 / 5G (model-dependent)                         |
| Network Features    | APN/VPDN, CHAP/PAP, ARP/Ethernet, static IP/DHCP         |
| Security            | Secure Boot, TPM2.0, TrustZone, firewall, VPN            |
| Cloud Management    | DeviceLive, HTTP/HTTPS/SSH remote management             |
| Data Acquisition    | Modbus RTU/TCP, EtherNet/IP, OPC UA, DNP3.0, BACnet, CNC |
| Open Platform       | Yocto/Linux with Debian 11 core root filesystem          |
| CPU/GPU             | Quad-core Cortex-A55 @ 2.0GHz / Mali-G52 2EE             |
| NPU                 | RKNN 1.0 TOPS (up to 26 TOPS with Hailo-8)               |

| Technical Indicator    | Specification                                                   |
|------------------------|-----------------------------------------------------------------|
| Memory/Storage         | 4GB / 16GB eMMC                                                 |
| Interfaces             | 4×GE, 4×RS-232/485/422 + 8×RS-485, CAN, DI/DO, USB, HDMI, mSATA |
| Power Input            | DC 12~48V (4W typical / 8W max)                                 |
| Dimensions (W × D × H) | 203 × 122 × 49 mm (7.99 × 4.8 × 1.93 in)                        |

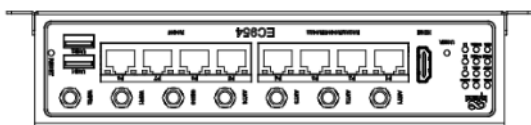
## 2. Product Dimensions



Front View



Side View



Interface Diagram

### Note:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and for reference only.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

### 3. Hardware Specifications

| Category/Parameter                   | Specification                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------|
| <b>Hardware Platform</b>             |                                                                                   |
| CPU                                  | Quad-core Cortex-A55 @ 2.0GHz                                                     |
| GPU                                  | Mali-G52 2EE                                                                      |
| NPU                                  | RKNN 1.0 TOPS (up to 26 TOPS with Hailo-8 module)                                 |
| RAM                                  | 4GB                                                                               |
| FLASH                                | 16GB eMMC                                                                         |
| <b>Connectivity &amp; Interfaces</b> |                                                                                   |
| Ethernet Ports                       | 4×10/100/1000 Mbps GE                                                             |
| I/O Ports                            | 4×DI + 4×DO                                                                       |
| Serial Ports                         | 4×RS-232/485/422 + 8×RS-485                                                       |
| CAN                                  | 2×CAN2.0A/B                                                                       |
| Buttons                              | Pinhole reset button                                                              |
| SIM Card Holders                     | Dual-SIM                                                                          |
| LED Indicators                       | 4G, SignalStrength (L1, L2, L3), SIM1, SIM2, User1, User2, PWR, STATUS, WARN, ERR |
| USB                                  | USB2.0 (2×Type-A + 1×Type-C)                                                      |
| microSD                              | Supports microSD                                                                  |
| Expansion Interfaces                 | 1×mSATA, B-Key 3042 (Hailo AI module)                                             |
| HDMI                                 | HDMI2.0                                                                           |

| Category/Parameter                      | Specification                            |
|-----------------------------------------|------------------------------------------|
| Wi-Fi                                   | STA, 802.11ac/a/b/g/n, 2.4G/5G dual band |
| Bluetooth                               | BLE 5.0                                  |
| GPS                                     | GPS/Beidou/GLONASS                       |
| <b>Power &amp; Power Consumption</b>    |                                          |
| Input Voltage                           | DC 12~48V                                |
| Power Interface                         | DC terminal input                        |
| Typical Value (OS Idle State)           | 4W                                       |
| Maximum Value (Full Load)               | 8W                                       |
| <b>Mechanical Specifications</b>        |                                          |
| Product Dimensions                      | 203 × 122 × 49 mm (7.99 × 4.8 × 1.93 in) |
| Mounting Method                         | DIN-rail / wall mounting                 |
| Protection Rating                       | IP30                                     |
| Enclosure & Heat Dissipation            | Metal housing, fanless design            |
| TPM                                     | TPM 2.0                                  |
| <b>Environment &amp; Certifications</b> |                                          |
| Storage Temperature                     | -40 °C ~ 85 °C (-40 °F ~ 185 °F)         |
| Operating Temperature                   | -20 °C ~ 70 °C (-4 °F ~ 158 °F)          |

| Category/Parameter       | Specification                                                                                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental Humidity   | 5~95% RH (non-condensing)                                                                                                                                                                                                                                            |
| Physical Characteristics | IEC60068-2-27 shock resistance<br>IEC60068-2-6 vibration resistance<br>IEC60068-2-32 drop resistance                                                                                                                                                                 |
| EMC Standard             | EN61000-4-2, level 3, Static<br>EN61000-4-3, level 3, Radiation Electric Field<br>EN61000-4-4, level 3, Pulsed Electric Field<br>EN61000-4-5, level 3, Surge<br>EN61000-4-6, level 3, Conducted Disturbance Immunity<br>EN61000-4-12, level 3, Shock Wave Resistance |
| Certifications           | CE, FCC, PTCRB, Verizon, AT&T                                                                                                                                                                                                                                        |

## 4. Software Specifications

| Category/Parameter      | Specification                        |
|-------------------------|--------------------------------------|
| <b>Operating System</b> |                                      |
| Operating System        | Yocto/Linux (Debian 11, Kernel 5.10) |
| File System             | Debian core root filesystem          |
| Package Manager         | Debian package manager               |
| <b>Network Features</b> |                                      |
| Network Type            | LTE Cat.4 (5G by model)              |
| Network Access          | APN, VPDN                            |
| Access Authentication   | CHAP/PAP                             |

| Category/Parameter                      | Specification                                                                                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WAN Protocols                           | Static IP, DHCP                                                                                                                                                              |
| LAN Protocols                           | ARP, Ethernet                                                                                                                                                                |
| <b>Security</b>                         |                                                                                                                                                                              |
| Secure Boot                             | Supported                                                                                                                                                                    |
| Trust Zone                              | Supported                                                                                                                                                                    |
| Network Security                        | Firewall                                                                                                                                                                     |
| Data Security                           | VPN                                                                                                                                                                          |
| <b>Reliability</b>                      |                                                                                                                                                                              |
| Link Detection                          | Multi-level link detection, auto-redial                                                                                                                                      |
| Built-in Watchdog                       | Device self-diagnosing, auto-recovers from operation faults                                                                                                                  |
| Backup Mechanism                        | Dual-SIM backup                                                                                                                                                              |
| Dual-SIM Switchover                     | Supported                                                                                                                                                                    |
| <b>Data Acquisition Protocols (DSA)</b> |                                                                                                                                                                              |
| Industrial Protocols                    | Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINSUDP, HostLink, PPI |
| Electricity Protocols                   | DLT645-2007, IEC101/104, DNP3.0                                                                                                                                              |
| Other Protocols                         | BACnet, CNC                                                                                                                                                                  |
| <b>Network Management</b>               |                                                                                                                                                                              |

| Category/Parameter   | Specification                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------|
| Upgrade Method       | Supports patent upgrade mechanism, local or remote firmware upgrade                                     |
| Configuration Method | WEB configuration                                                                                       |
| Log Functions        | Support local system logs, remote logs, and important log power-off preservation                        |
| Remote Management    | DeviceLive / HTTP / HTTPS / SSH remote access                                                           |
| DeviceLive Cloud     | Supports cloud-based parameter configuration, container management, application and firmware management |

## 5. Ordering Information

### Model Rule

**Model code:** EC954-<WMNN>-B-[XY]-[Z]

<WMNN>: Cellular Type & Frequency Band (mandatory)

**Standard configuration (all models):** Memory/Storage 4GB/16GB · Ethernet/Serial 4×1000Mbps; 4×RS-232/485/422 + 4×RS-485 · Wi-Fi/BT/GPS/CAN/TPM & I/O included

[XY] (Optional): AI Expansion Module

[Z] (Optional): OS option



## Model List

| Model        | Region           | Cellular Type & Frequency Band                                                                                                    |
|--------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| EC954-LQA8-B | China            | LTE Cat.4<br>LTE-FDD B1/B3/B5/B8; LTE-TDD<br>B34/B38/B39/B40/B41; WCDMA B1/B8;<br>TD-SCDMA B34/B39; CDMA BC0; GSM<br>900/1800MHz  |
| EC954-NRQ1-B | China            | 5G NR<br>NSA n78/n79; SA<br>n1/n3/n5/n8/n28/n41/n77/n78/n79; LTE-<br>FDD B1/B3/B5/B8; LTE-TDD<br>B34/B38/B39/B40/B41; WCDMA B1/B8 |
| EC954-FQ58-B | EMEA             | LTE Cat.4<br>LTE-FDD B1/B3/B7/B8/B20/B28A; LTE-<br>TDD B38/B40/B41; WCDMA B1/B8; GSM<br>B3/B8                                     |
| EC954-FQ38-B | North<br>America | LTE Cat.4<br>LTE-FDD B2/B4/B5/B12/B13/B14/B66/B71;<br>WCDMA B2/B4/B5                                                              |
| EC954-EN00-B | Global           | No Cellular                                                                                                                       |

## AI Expansion Module (Optional)

| [XY] P/N Code | Feature                   |
|---------------|---------------------------|
| —             | None                      |
| H8            | Hailo-8, M.2 Key B+M 2280 |

## OS Option (Optional)

| [Z] P/N Code | Feature         |
|--------------|-----------------|
| —            | IEOS (default)  |
| D            | Debian Linux OS |

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